Math 534H - Spring 2026 - Lecture 5

A different way of solving $u_x + 3u_y - u = 0$ using the <u>COORDINATE METHOD</u> instead.

Let's do it more generally $\boxed{a u_x + b u_y + c u = 0}$ (*)

(in our example $a=1$, $b=3$, $c=-1$)

<u>Let</u>
$$\begin{cases} \xi = ax + by = \vec{v}\cdot(x,y) \\ \eta = bx - ay = \vec{v}^{\perp}\cdot(x,y) \end{cases}$$

$\vec{v} = (a,b)$

$\vec{v}^{\perp} = (b,-a)$

$$u_x = a\cdot u_\xi + b u_\eta$$
$$u_y = b u_\xi - a u_\eta$$

$$\underbrace{\frac{\partial u}{\partial x}}_{u_x} = \underbrace{\frac{\partial u}{\partial \xi}}_{u_\xi}\cdot\underbrace{\frac{\partial \xi}{\partial x}}_{a} + \underbrace{\frac{\partial u}{\partial \eta}}_{u_\eta}\cdot\underbrace{\frac{\partial \eta}{\partial x}}_{b}.$$

Now coming back to ⊛ and substituting u_x, u_y we get that as a function of ξ and η ⊛ becomes

$$\underbrace{(a^2 u_\xi \cancel{+ ab u_\eta})}_{a u_x} + \underbrace{(b^2 u_\eta \cancel{- ba u_\xi})}_{b u_y} + cu = 0.$$

$$(a^2+b^2)\, u_\xi + cu = 0 \quad \overset{a \neq 0 \quad b \neq 0}{\Longleftrightarrow}$$

$$u_\xi = -\frac{c}{(a^2+b^2)} \cdot u$$

Wherever $u(\xi,\eta) \neq 0$ we can divide by u (Domain of validity)

$$\frac{u_\xi}{u} = -\frac{c}{(a^2+b^2)}$$

$$\partial_\xi(\ln|u|) = -\frac{c}{(a^2+b^2)}$$

integrate both sides wrt ξ $\Longrightarrow$

$$\ln|u| = -\frac{c}{(a^2+b^2)}\cdot\xi + \underbrace{g(\eta)}_{\text{"constant" is a function of } \eta \text{ b/c } \partial_\xi g(\eta) = 0.}$$

where g is any differentiable function of 1 variable.

Hence exponentiating both sides we get.

$$u(\xi,\eta) = e^{\left[-\frac{c}{(a^2+b^2)}\cdot\xi + g(\eta)\right]} \qquad e^{g(\eta)} = \phi(\eta)$$

$$= e^{-\frac{c}{a^2+b^2}\cdot\xi}\cdot\phi(\eta)$$ where ϕ is any differentiable function of 1 variable. Going back to the variables (x,y); we get

$$\boxed{u(x,y) = \phi(bx-ay)\cdot e^{-\frac{c(ax+by)}{a^2+b^2}}}$$

$a=1 \quad b=3 \quad c=-1$

$$-\frac{c(ax+by)}{a^2+b^2} = \frac{x+3y}{10} \quad ?$$

<u>Class Hmwk:</u> Check this indeed solves

$$a u_x + b u_y + c u = 0.$$

<u>Next time:</u> How do we solve constant coefficient <u>general inhomogeneous</u> equation:

$$\begin{cases} u_t + b u_x = \underline{F(x,t)} \\ u(x,0) = g(x) \end{cases} \quad (x,t) \in \mathbb{R} \times \mathbb{R}.$$

with initial condition

<u>MAIN HOMEWORK</u> 1: <u>Read section 1.4</u>: Auxiliary conditions (initial condition and/or boundary conditions)

- Dirichlet Boundary Conditions
- Neumann " "
- Robin boundary conditions

2 Read Section 1.5 on meaning of WELL POSEDNESS (existence, uniqueness, stability).